LECTURE 05

Motion begins with a reference

Frame of Reference, Position, Distance and Displacement

Class 11 Physics • CBSE + NEET Foundation • 2 hours

Observe → Define → Represent → Compare → Apply

Chapter 2 • Motion in a Straight Line

What we will accomplish

- Decide whether an object is at rest or in motion relative to an observer
- Choose an origin, direction and coordinate to describe position
- Calculate distance travelled and displacement
- Distinguish scalar distance from vector displacement
- Solve CBSE numericals and NEET-style conceptual questions

Success check: explain why distance can be non-zero while displacement is zero.

Recall before we start

1. What makes a physical quantity complete?
2. What is the SI base unit of length?
3. Is speed a base or derived quantity?
4. What does a negative sign usually indicate in one-dimensional motion?

At rest or in motion?

TRAIN FRAME

Passenger sitting inside a moving train

Result: at rest relative to the train

GROUND FRAME

Observer standing on the platform

Result: passenger is moving

KEY IDEA **Same passenger • different observers • different descriptions**

Rest and motion are relative

REST

Position does not change with time relative to the chosen reference frame.

MOTION

Position changes with time relative to the chosen reference frame.

Frame of reference

Reference frame = origin + axes + clock

A system used to specify position and motion

Origin O

Where position is measured from

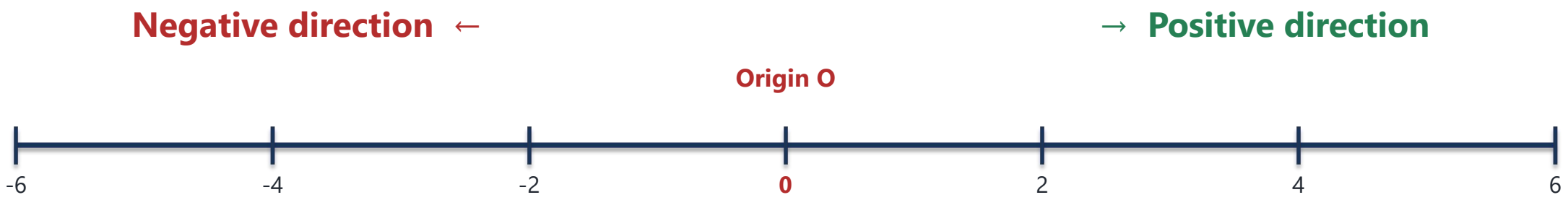
Direction / axes

Which way is taken as positive

Clock

When each observation is made

Build a one-dimensional reference frame



Coordinate tells location relative to the selected origin—not absolute location.

Position depends on the chosen origin

Observer A

Origin is 2 m left of the tree

Position of tree: x = +2 m

Observer B

Origin is 5 m right of the tree

Position of tree: x = −5 m

Different coordinates can describe the same physical location.

Position

Position specifies where an object is located relative to the chosen origin.

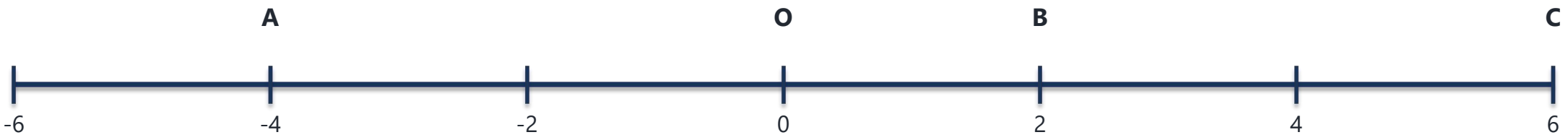
In one-dimensional motion, position is represented by the coordinate x.



A: x = −4 m **B: x = +3 m**

SI unit: metre (m) • **Dimension: [L]**

Reading position on a number line



- State the position of A, B and C.
- Which object is farthest from the origin?
- What is the separation between A and B?

Initial and final position

Initial position

$x_i = -2$ m

Final position

$x^f = +5$ m

Change in position = $x^f - x_i$

Always subtract the initial position from the final position

Checkpoint 1

- A student sits inside a moving bus. State one frame in which the student is at rest and one in which the student is moving.
- An object is 3 m left of the origin. Write its position.
- Does negative position necessarily mean the object is moving in the negative direction? Explain.

PAIR • EXPLAIN • CORRECT

RESET

Short break

5 minutes

Return ready to compare distance and displacement.

Recall during the break: origin • direction • position

DEFINITION

Distance travelled

Total length of the actual path followed by an object.

Distance = sum of lengths of all path segments

Scalar • path-dependent • never negative • SI unit: m

Common error: distance is not always |final position − initial position|.

DEFINITION

Displacement

Change in position from the initial point to the final point.

$$\Delta x = x^f - x_i$$

Vector • path-independent • can be positive, negative or zero

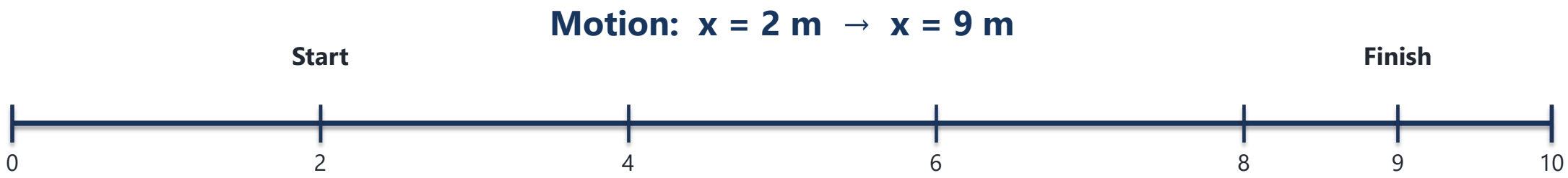
Magnitude of displacement = $|\Delta x|$ and is always non-negative.

COMPARE

Distance versus displacement

Feature	Distance	Displacement
Meaning	Actual path length	Change in position
Nature	Scalar	Vector
Path	Depends on path	Only endpoints matter
Possible value	Zero or positive	Positive, negative or zero
SI unit	metre (m)	metre (m)

Worked example 1: straight path



Distance = 7 m

Δx = 9 − 2 = +7 m

Worked example 2: reversal of direction

Path: 2 m → 8 m → −3 m



Distance = 6 + 11 = 17 m

Δx = −3 − 2 = −5 m

Worked example 3: round trip

Runner completes one 400 m lap and returns to the starting point.

Initial position = Final position

Distance = 400 m

Displacement = 0

The fundamental relation

Distance ≥ |Displacement|

For every journey

Equal

Straight-line motion
without reversing direction

Distance is greater

Curved path, detour
or reversal of direction

What does the sign of displacement mean?

$\Delta x > 0$
Net change in positive direction

$\Delta x < 0$
Net change in negative direction

$\Delta x = 0$
Initial and final positions coincide

APPLICATION

Real-life instruments tell different stories

ODOMETER

Records total path length travelled by the vehicle

DISTANCE

MAP / GPS

Can compare start and finish with direction

DISPLACEMENT

SOLVE ON THE BOARD

Guided practice

- A moves from $x = -5$ m to $x = +4$ m, then returns to $x = -2$ m. Find distance and displacement.
- A child walks 8 m east and then 8 m west. Find distance and displacement.
- An object moves directly from $x = +12$ m to $x = -8$ m. Find distance and displacement.

Show: path segments • formula • substitution • unit • direction

NEET concept check

1. Distance equals magnitude of displacement when motion is:

A. circular B. straight without reversal C. random D. always

2. After one complete revolution on a circular track, displacement is:

A. circumference B. radius C. zero D. diameter

3. Which quantity changes if only the origin is shifted?

A. distance travelled B. position coordinate C. path length D. elapsed time

Choose first. Justify second.

CBSE written-answer practice

Question: Distinguish between distance and displacement with one suitable example.

1 mark
Correct definitions

1 mark
Scalar/vector and path dependence

1 mark
Example with values and units

Common errors to eliminate

- Treating distance and displacement as identical
- Forgetting: displacement = final position − initial position
- Using a negative sign for distance
- Assuming zero displacement means no motion occurred
- Writing a numerical answer without unit or direction

Exit test

- What must be specified before calling an object at rest or in motion?
- Write the position of an object 6 m left of the origin.
- A moves 8 m east and 6 m west. Find distance and displacement.
- What is displacement after returning to the starting point?
- When does distance equal magnitude of displacement?

Takeaway and daily practice

Frame origin + direction + clock

Position location relative to origin

Distance total actual path length

Displacement $\Delta x = x^f - x_i$

Next: average speed, average velocity and motion graphs